



A serial-position curve in high-performance darts: The effect of visuomotor calibration on throwing accuracy

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Abstract

The aim of the present research was to test if the fine-tuning of skilled motor actions benefits from proximate previous actions via a visuomotor calibration process. In professional darts, each player cycles through different activities: three darts are thrown with a rather smooth sequence of movements, the darts are retrieved from the dartboard, the other player throws his or her darts and retrieves them, the next three darts are thrown, retrieved, etc. We hypothesized that these cycles give rise to a serial-position curve for the precision of darts as a result of a particular kind of warm-up decrement. Even though the interruptions of actually throwing darts are only in the order of seconds, walking away from the throw line should lead to a loss of fine-tuning of the calibration of movement parameters with respect to targets defined in the external frame of reference of the dartboard. For the players of the 2017 Professional Darts Corporation World Darts Championship ($N=36,168$ scores) we confirmed that the first dart of a series of three is indeed less accurate than the subsequent two. This warm-up decrement is particularly pronounced for vertical errors, for which the relation to movement parameters is more complex than for horizontal errors. Fine-tuning of visuomotor calibration is a neglected facet of warm-up that is also important for various other sports such as tennis, basketball, handball, and football.

Introduction

Playing darts involves a cycle of different activities. In a match between two players, three darts are thrown by the one player and retrieved from the dartboard. Then three darts are thrown by the other player (which is referred to as a throw in darts terminology) and retrieved from the dartboard. Thus, each player throws a series of three darts one after the other (an individual dart throw is referred to as dart in darts terminology) before this rather smooth sequence of movements is interrupted by retrieving his or her darts from the dartboard and by the other player throwing his or her darts and retrieving them. This cycle is repeated for each player until one of them reaches a predefined score. Pertinent to the present research, there is reason to hypothesize that this

cycle of throwing, being interrupted, throwing, etc., might go along with periodic variations of the precision of the darts. Here we show the existence of such variations in high-performance darts, specifically in the performance of the players of the 2017 Professional Darts Corporation World Darts Championship.

Among the classic findings of motor-skills research is the warm-up decrement (Adams, 1961). This is a decline of performance during a break that is rapidly compensated when the motor task is resumed. Although the warm-up decrement has been known for a long time, only little research effort has been devoted to understanding its origin. In the sixth edition of a popular textbook the rather nonspecific “set hypothesis” still appears as the predominant explanation of the effect (Schmidt, Lee, Winstein, Wulf, & Zelaznik, 2019, p. 419), just as in the first edition (Schmidt, 1982, p. 624). The term “set” refers to task-specific adjustments of arousal, attention, and other mental and bodily states that have the potential to affect performance. However, concerning high-performance darts, it seems quite unlikely that task-specific adjustments of arousal, attention, and other bodily and mental states have the potential to account for performance variations in the short cycles typical for darts: cycle durations are in the order of seconds rather than in the order of minutes as the

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breaks in studies of warm-up decrement are. Nevertheless, there is reason to hypothesize that a specific loss of adjustment could give rise to a warm-up decrement for the first dart in the row of three: this is the loss of fine-tuning of the calibration of movement parameters with respect to the external frame of reference defined by the dartboard. While it has been argued that a loss of fine-tuning of well learned skills can be observed in professional athletes and, therefore, highly skilled athletes (e.g., golf players, baseball players, etc.) need to calibrate their perceptual-motor system before competing (Ajemian, D'Ausilio, Moorman, & Bizzi, 2010), we are not aware of any data from professional sport competitions supporting this line of argumentation. Finding a warm-up decrement within the short time scale of seconds in the highest level of darts players would, therefore, provide strong evidence for the perceptual-motor calibration hypothesis of the warm-up decrement as it seems unlikely that adjustments of arousal, attention, and other bodily and mental states proposed in the “set hypothesis” could explain these short time fluctuations.

The precision of throwing darts depends on movement parameters that are somewhat different for the horizontal and the vertical axis (cf. Smeets, Frens, & Brenner, 2002). Darts are thrown in overarm fashion with a forward movement of the arm, mainly an elbow extension. This movement is straight in the horizontal plane and slightly curved in the vertical plane. The horizontal precision of the throw depends primarily on the direction of the straight horizontal component of the arm movement, with the horizontal component of the flight of the dart being roughly a continuation of the horizontal component of the arm movement along a straight line. The vertical precision depends on the position of the dart together with its vertical direction and velocity at the moment of release, and therefore, on the timing of release. One strategy that can be used to enhance the vertical precision is the choice of equifinal movements. These are movements during which direction and velocity co-vary in such a way that initial direction and velocity of the dart are appropriate for reaching the target across a considerable range of release times (Müller & Loosch, 1999). This kind of strategy has been found in some expert players, but not in others for whom the release timing was quite precise (Nasu, Matsuo, & Kadota, 2014). Thus, for the horizontal error of the impact of the dart on the board it is mainly the precision of the horizontal direction of arm movement that is crucial, whereas for the vertical error mainly the precision of the position, the vertical direction and the velocity of the dart at release matter, that is, the temporal precision of the grip opening during the forward movement of the hand.

For a dart to hit a desired target, movement parameters, which are defined in a body-related frame of reference, have to be calibrated with respect to an external frame of reference. The link between these different frames of reference is

generally established by vision. Without vision, the frames of reference can drift apart, which results in gross movement errors (e.g., Admiraal, Keijsers, & Gielen, 2004; Medendorp, Van Asselt, & Gielen, 1999). An example is pointing at memorized targets in microgravity, where due to the absence of gravity the orientation of the body in space and thus its relation to the environment becomes uncertain fairly rapidly. As a consequence pointing becomes quite inaccurate (Watt, 1997). Another example is the phenomenon of unintended acceleration of automobiles with automatic transmission, which results from pressing the accelerator rather than the brake at the time the gear selector is moved to D after the driver has just entered the car (Schmidt, 1989). The foot is invisible, so the foot movement in a certain direction can reach the wrong pedal when the body-related and external frames of reference are misaligned. This is less likely to occur in cars with manual transmissions as the left foot operates the transmission and hereby gives a frame of reference for movements of the right foot.

In throwing darts or other objects the target can be seen. The link between body-related and external frames of reference is established by fixating the target (probably in combination with a common axis defined, e.g., by the vertical). More specifically, fixation links a position in the external frame of reference to the orientation of the eyes. As a consequence, the fixated position is not only specified in the external frame of reference, but also in body-related frames (vertical and horizontal orientations of the eyes relative to the head, vertical and horizontal orientations of the head on the shoulder, etc.) Thus, fixation establishes a kind of “entrance” of a position in external space into representations of the body and the configuration of its segments. In fact, fixating the target facilitates accurate throwing (cf. Vine, Moore, & Wilson, 2014).

Even when external and bodily frames of reference are linked in principle, the computations required for an overarm throw suited to hit the fixated target are complex and require transformations between various frames. This is evident when one considers the transformations involved from muscle innervation to movement kinematics to the flight of the dart (cf. Heuer & Massen, 2013). The mastery of such transformations is generally ascribed to the computational use of their internal representations which are known as internal models (e.g., Wolpert & Kawato, 1998). A part of these transformations, ranging from muscle innervation to kinematics of the movement, is defined in body-related frames of reference. We refer to the total of these transformations as the “motor transformation”. In throwing darts, movement parameters such as horizontal and vertical directions as well as position and velocity of the hand and thus of the dart at the moment of release, which are defined in body-related frames of reference, are further transformed into positions of impact of the dart on the dartboard, which

are defined in an external frame of reference. We refer to this transformation as the “visuomotor transformation”. The representation of this type of transformation between movement parameters and movement outcomes is also known as (recall) schema (Schmidt, 1975).

The transformations involved in precise throwing are variable—they change when bodily characteristics vary and, more frequently, when we use optical devices or different tools (e.g., when throwing a ball with a lacrosse stick). This requires flexibility in getting along with different transformations. Perhaps the most impressive evidence of this flexibility is adaptation to a lateral displacement of visual targets when looking through wedge prisms. Though most of the studies show adaptation for reaching, it has also been observed for throwing (Martin, Keating, Goodkin, Bastian, & Thach, 1996a, b). The flexibility required for the mastery of variable transformations and the ever-present noise in neural systems suggest that the relation between movement parameters and movement outcomes is noisy to some degree. Hence, and of particular importance to the present research, visuomotor calibration—the calibration of movement parameters with respect to targets seen in the environment—is unlikely to be perfect and likely to require fine-tuning upon every displacement of one of the involved frames of reference relative to the other frames.

Returning to the initially mentioned cycles of high-performance darts, players move away from the oche (or throw line) to retrieve their darts from the board and step aside. For the next throw of three darts they have to re-adjust their body placement and configuration for a fine-tuning of the visuomotor calibration. Any information on the exactness of this re-adjustment and thus the proper visuomotor calibration is provided only by the first dart. In this respect the first dart of a series of three could serve a similar fine-tuning function as the often seen preliminary gentle hammer blow that precedes the effective powerful blow when hitting a nail. However, the first dart is not preliminary, but scored. These considerations led us to expect that the first dart of a throw should be slightly less accurate than the subsequent darts.

Method

The database comprised 36,168 scores, almost all scores in matches played during the 2017 PDC World Darts Championship. The score of every single dart thrown by a certain player was recorded as a function of its sequential order in a leg. A sub-sample of video recordings of four matches was randomly selected and observed and coded by two experienced research assistants. The results showed a high reliability (coefficients of agreement of 1.0 for points scored).

Data were extracted from televised video recordings of almost all matches during the 2017 Professional Darts

Corporation (PDC) World Darts Championship. These were 29 of 32 matches in round one, all 16 matches in round two, 8 in round three, 4 quarterfinals, 2 semifinals, and 1 final. Only for 3 matches of round one, video footage was missing. The database contains 36,168 scores of 61 of the 64 players.

The Bristle dartboard, as used during the 2017 PDC World Darts Championship and illustrated in Fig. 1, has a diameter of 340 mm measured for the outer ring of the outer band. The outer ring of the inner band has a diameter of 214 mm. The two bands that define Doubles and Triples are 8 mm wide. The outer central ring has a diameter of 31.8 mm and the inner central ring (bull’s eye) of 12.7 mm.

Figure 1 shows a Bristle dartboard as used during the tournament which illustrates the scores assigned to each segment. The outer band defines a Double and the inner band a Triple. The outer central ring scores 25, and the inner central ring scores 50. Thus, the maximal scores for each dart are the Triple 20, the Triple 19, the Triple 18, and the Triple 17, followed by the inner central ring (bull’s eye).

The rules for the 2017 world championship can be found at <http://www.thedra.co.uk/wp-content/uploads/2015/01/DRA-Rules-final-140115.pdf>. The tournament was run on a knockout basis: two players played one-on-one against each other a variant of darts called 501 with “double out”. Here the initial score of each player is 501. The players take turns to throw three darts, called a throw, on a standardized dart board. The scores are subtracted from the total score until the end of the so-called leg. The leg is won by the player whose score reaches zero first. However, zero has to be reached with a Double (e.g., when a player has a remaining score of 16, he or she must hit Double 8 to win the leg). If a dart reduces the total score of a player below zero, exactly to one point, or to zero without hitting a Double the throw is invalid and the total score is reset to the value prior to the throw. A player had to win 3 legs (best of 5) to win

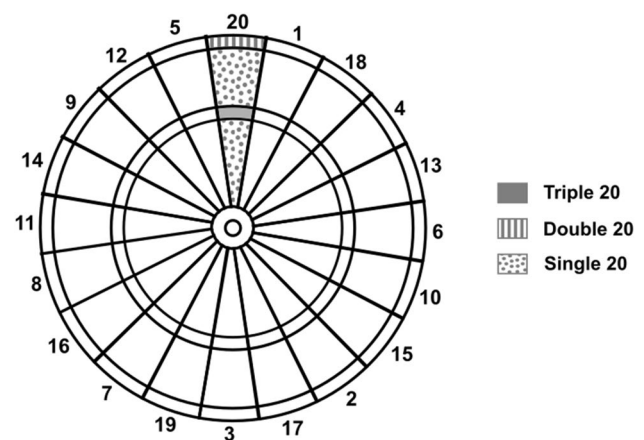


Fig. 1 Illustration of a Bristle dartboard with the segment scoring 20 marked (not shown are the colours of the board)

one set. The first round was best of 5 sets (first to win 3), the second and third round was best of seven sets (first to win 4), the quarterfinals were best of 9 sets (first to win 5), the semifinals were best of 11 sets (first to win 6), and the final was best of 13 sets (first to win 7).

Results

Analysis of scores

Figure 2a shows the mean score of each dart as a function of its serial position within each throw for the first five throws, averaged across all players and all legs. The first five throws were chosen for illustration purposes, because empirically the fifth is the first throw where a noticeably reduced number of darts and a wide use of segments by the players can be observed. For the first three throws, each of three darts, the number was maximal because nine is the minimal number of darts to win a leg.

There are two conspicuous features of this serial-position curve. First, in the course of a leg the mean scores decline. This is a consequence of the rules as players typically have to maximize their score only early in a leg, but finish the leg with smaller scores to exactly reach zero by hitting a Double with their final dart. This change of strategy within a leg is apparent from the frequency distributions of the scores for the first dart of throw 1 (Fig. 2b) and the last dart of throw 5 (Fig. 2c).

The first dart seems to be essentially always directed at the Triple 20. It was hit in several instances, but more frequently it was missed. The most frequent score of 20 is such a miss, a vertical error relative to the target of Triple 20, and the Triple 5 and Triple 1 are horizontal errors (cf. Fig. 1). For the scores of 1 and 5 both vertical and horizontal errors are combined. Other scores were only one 0 and one Triple 19 (not visible in Fig. 2b because of the scale of the ordinate). For the third dart of throw 5 the distribution of scores is much wider. Depending on the current total score the players are in need to aim for specific segments which is reflected in the data. Moreover, a significant proportion of darts does not achieve a score at all. This usually reflects players' unsuccessful targeting of a Double to finish the leg, using the strategy of initially aiming further outside as a miss outside gives them another chance of targeting the same Double again.

The second, more relevant, conspicuous feature of the serial-position curve of Fig. 2a is the lower score of dart 1 compared to darts 2 and 3 of a throw, which is seen for the first three throws, but not for the later ones. This is a remarkable result, knowing that thrown darts can constitute an obstacle for consecutive darts as these can (partly) block target areas. Thus, players could choose to change the

segment they aim at or to perform a risky throw. The strategic choice of changing a targeted segment complicates our analyses. However, for the first six darts of a player in a leg we can assume that they are aimed at the maximal scores possible. After three darts the minimum total score is 321, and after 6 darts it is 141. Only thereafter the player has to make strategic choices other than maximizing the score of each dart.

Analysis of accuracy

As described above, the precise segment a player is aiming at is not known. However, two goals for the first six darts can be inferred without doubt: first, the player aims at a Triple, and second, he or she aims at a segment with a high-score (i.e., 20, 19, 18 or 17; cf. Fig. 1). Therefore, we classified each dart according to its score into one of four categories:

- (a) "Hit": the dart hits one of the targets Triple-20, Triple-19, Triple-18, or Triple-17.
- (b) "Vertical error": the dart hits a segment with a high-score, but misses the Triple (Single-20, Single-19, Single-18, or Single-17). Note that the high-score segments on the dartboard have an approximately vertical orientation so that a vertical error in aiming at the Triple will result in a Single.
- (c) "Horizontal error": the dart hits a Triple, but misses the high-score segment; in the database only Triple-5, Triple-1, Triple-4, Triple-7, and Triple-3 occurred, which are adjacent to the high-score segments 20, 19, and 18. Note that the Triple band in the high-score segments and the adjacent ones has an approximately horizontal orientation.
- (d) "Other errors": all other scores, including combinations of vertical and horizontal errors.

For each player we determined the percentage of darts in each of these categories for each of the first 6 darts of all legs. The number of legs varied between players, ranging from 10 (first-round loser Dragutin Horvat) to 169 (runner-up Gary Anderson). Figure 3 shows the mean percentages of hits and the different types of errors.

From Fig. 3 it is apparent that the percentage of hits is smaller for the first dart of a throw than for the second and third dart. Consistent with this difference in the percentage of hits, the percentage of vertical errors is larger for the first dart of a throw. The other two types of errors are much less frequent, and the tendency of being more frequent for the first dart of a throw than for the subsequent ones is less pronounced or even absent. For the statistical analysis we used non-parametric tests because the Shapiro–Wilk test indicated significant deviations from normality for some of the distributions, in particular for horizontal errors and

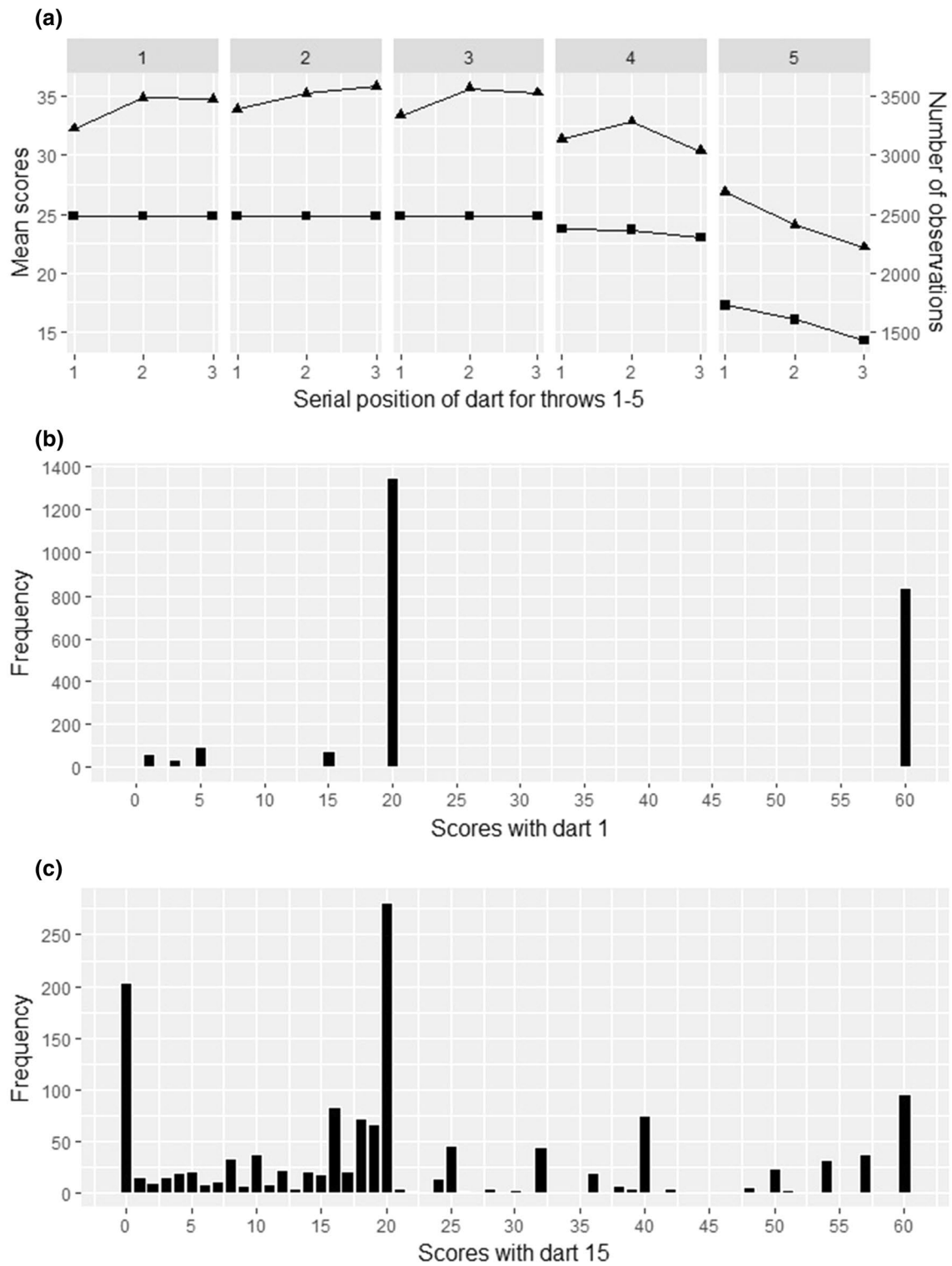


Fig. 2 **a** Mean scores (filled triangles) as a function of serial position of the dart for the first five throws (a throw is a series of 3 darts) and number of observations (filled square) for each serial position, **b** Fre-

quencies of scores for dart 1, the first dart of the first throw, **c** Frequencies of scores for dart 15, the third dart of the fifth throw

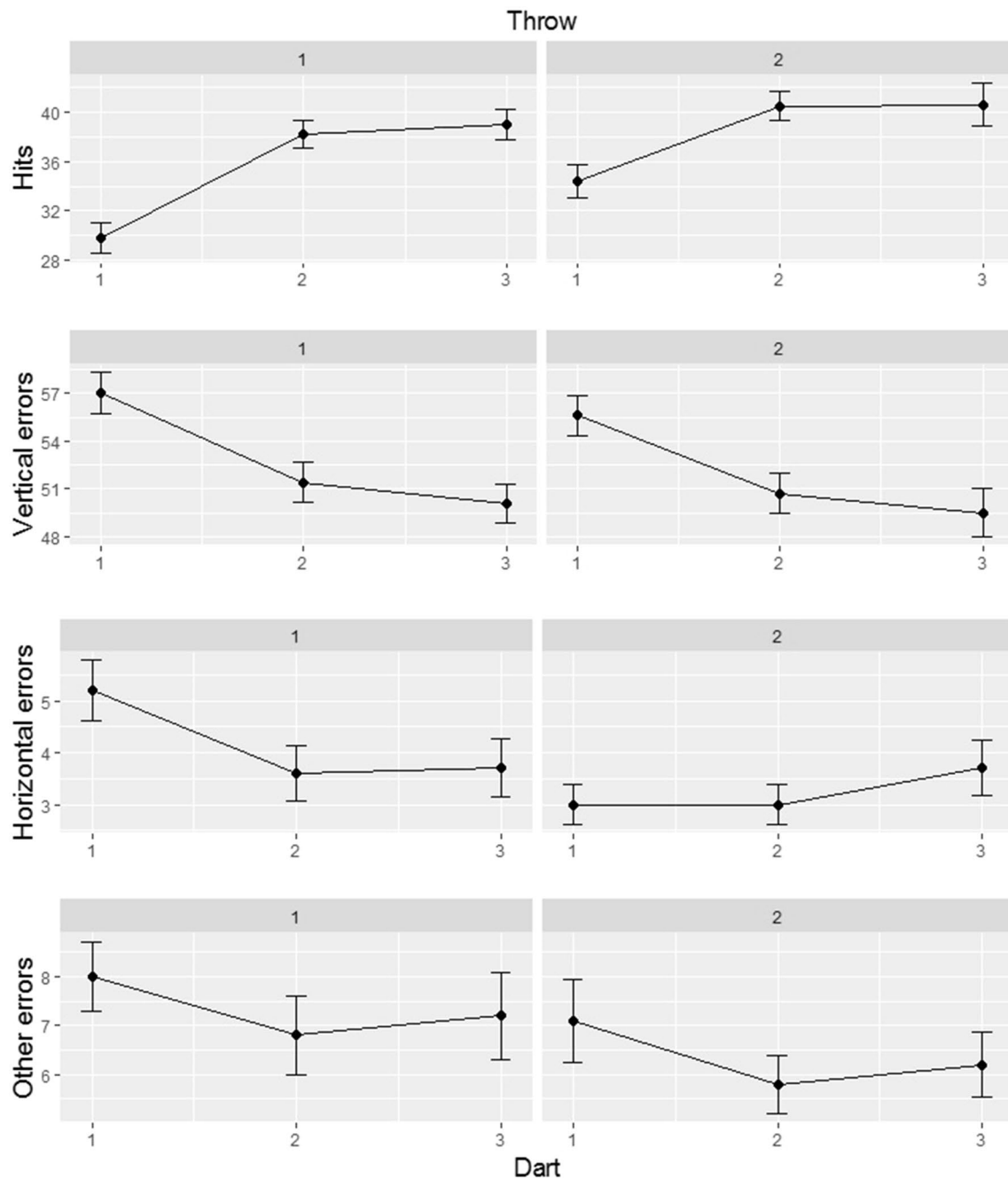


Fig. 3 Mean percentages (with standard errors) of hits, vertical errors, horizontal errors, and other errors for each dart of the first two throws

other errors. As indicated by Friedman tests, for the first throw the difference between the three darts was significant for hits ($\chi^2_2 = 31.56, p < 0.001$), for vertical errors ($\chi^2_2 = 10.57, p < 0.01$), and for horizontal errors ($\chi^2_2 = 10.78, p < 0.01$), but not for other errors ($\chi^2_2 = 1.25, p = 0.535$). For the second throw, the difference between the three darts was significant for hits ($\chi^2_2 = 15.46, p < 0.001$) and

for vertical errors ($\chi^2_2 = 14.26, p < 0.001$), but not for horizontal errors ($\chi^2_2 = 0.15, p = 0.926$) and for other errors ($\chi^2_2 = 0.01, p = 0.994$). A series of Wilcoxon signed-rank tests with Bonferroni correction was performed between pairs of serial positions within each throw, the results of which are presented in Table 1. These show that the “deviant” serial position in each throw was in fact the first one.

Table 1 *p* values for Friedman tests and Bonferroni-corrected Wilcoxon post hoc test

Throw	Category	Friedman tests		Wilcoxon tests	
		Dart 1 vs 2 vs 3	Dart 1 vs 2	Dart 1 vs 3	Dart 2 vs 3
1	Hit	<0.001*	<0.001**	<0.001**	0.816
	Vertical error	0.005*	0.002**	<0.001**	0.270
	Horizontal error	0.005*	0.005**	0.021	0.934
	Other error	0.535	–	–	–
2	Hit	<0.001*	0.002**	0.001**	0.419
	Vertical error	<0.001*	0.006**	<0.001**	0.253
	Horizontal error	0.926	–	–	–
	Other error	0.994	–	–	–

*Significant at 5% level

**Significant at 5% level using Bonferroni correction

Discussion

The goal of the present study was to test the hypothesis of a cyclic variation of performance in high-performance darts. The hypothesis was based on the assumption of a rapid warm-up decrement during retrieval of the darts from the dartboard and the waiting until the other player had finished his or her throw and retrieved his or her darts. It can reasonably be tested only for the first two throws (i.e., six darts) of a leg because only here one can assume that different players aim for a limited set of score-maximizing targets. We found support for the hypothesis in that the proportion of hits was smaller for the first darts of the first and second throw, whereas the proportion of errors was larger. This variation in accuracy was accompanied by a variation of scores of each dart.

The notion of a warm-up decrement is well-defined regarding its formal definition, but ill-defined with respect to the specific origins of the performance decrement during a break and the rapid recovery after the break. In high-performance darts, the “break” is a short interruption of throwing a series of darts that lasts for several seconds. We conjecture that the main source of the warm-up decrement is leaving the oche and thereby displacing the bodily frames of reference relative to the external frame. The main source of the subsequent warm up is then the fine-tuning of the visuomotor calibration after the feedback provided by the first dart of a throw. This fine-tuning is likely a delicate process. Therefore, it seems feasible that this effect could be hidden or less pronounced in less-skilled players due to the variability of performance. Future research might want to investigate into warm-up decrements as a function of experience and expertise (see Ajemian et al., 2010 for a detailed description of why this warm-up decrement might only be limited to highly skilled athletes).

What are the adjustments after the first dart of a throw that serve the fine-tuning of the visuomotor calibration? A well-founded answer to this question is beyond the scope of

the present study. However, to obtain first indications of the nature of the fine-tuning we distinguished between vertical and horizontal errors. Vertical errors were more than four times as frequent as all other errors together. A part of this difference, if not all, can be attributed to the different vertical and horizontal extents of the Triple, which are 8 mm and about 32 mm for the Bristle dartboard. However, a part could also be related to a larger scatter of vertical deviations from the target than of horizontal deviations. For vertical errors the improvement from the first to the second dart was consistent and conspicuous, whereas for the other error types it was less conspicuous and not always present. Although this could be a consequence of the overall smaller error percentage, we tentatively suggest that primarily the vertical accuracy suffers from the warm-up decrement and profits from the fine-tuning after the first dart of a throw. This would be consistent with the more complex motor control required for vertical accuracy than for horizontal accuracy. We are aware that this is speculative at present and acknowledge that future research is required to scrutinize the explanatory range and boundary conditions of the perceptual-motor calibration hypothesis of the warm-up decrement.

What is the practical relevance of the warm-up decrement in high-performance darts? This can be estimated from the mean scores per dart, as shown in Fig. 2a, when the difference between the first dart of each throw and the second and third dart is related to the mean scores of the first six darts of players who reached different stages of the tournament. These are listed in Table 2. Notably, the difference in mean scores for darts 1 and 2 of the first throw was 2.63 and for darts 1 and 3 it was 2.50. These differences are larger than the difference in mean scores of tournament winner Michael van Gerwen compared to the semifinalists Peter Wright and Raymond van Barneveld. Knowing that the difference in prize money between the winner and the semifinalists was 270.000 £ for the World Championship 2016/2017, the practical relevance of the warm-up decrement in professional darts can hardly be denied.

Table 2 Mean scores per dart of the first two throws of a leg by players achieving a specific tournament progress

Tournament progress	Mean	Difference to winner
Winner (van Gerwen)	39.07	0.00
Runner-up (Anderson)	38.92	0.15
Semifinal	36.94	2.14
Quarterfinal	34.81	4.26
Last 16	34.14	4.93
Last 32	33.29	5.78
Last 64	31.68	7.39

Given numerous anecdotal observations in professional sports (Ajemian et al., 2010) that could be attributed to expert athletes' awareness of warm-up decrements—e.g. tennis players playing several shots after a rain break, darts players taking several throws before a match starts, or handball and football players performing several passes after the half-time break—it is surprising that hardly any research has focused on the processes of visuomotor calibration during warm-up which might be engaged by the example activities mentioned. Previous literature has emphasized potential benefits of warming-up in sports in terms of temperature-related mechanisms (e.g., decreased stiffness, increased nerve-conduction rate, altered force–velocity relationship, increased anaerobic energy provision and increased thermoregulatory strain), or cardio-vascular (elevation of baseline oxygen consumption, $\dot{V}O_2$) and psychological preparation (Bishop, 2003a, b for reviews). Hence, the present study is also intended to inspire future research on visuomotor calibration during warm-up, which has the potential to inform evidence-based practice in competitive sports.

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Compliance with ethical standards

Conflict of interest All authors declare to have no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

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Data availability The data analyzed within the current study are available from the corresponding author on reasonable request.

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